

NUMERICAL ANALYSIS

PROBLEM SET 1

ROB TOMPKINS

We begin by noting that all of the files referenced in this document are available at: <https://github.com/chtompkins>

1. The value of π is given by the alternating series:

$$\pi = 4 \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = 4 \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots \right)$$

(a) Use this series to construct an algorithm to approximate the value of π to within a specified tolerance, ε . The algorithm should be general to allow for any ε . Provide a theoretical upperbound ϵ on the error of your approximation. Hint. It may be useful to recall known properties of the partial sums of alternating series.

(b) Run your algorithm with a tolerance value of $\varepsilon = 5 \times 10^{-3}$ and show that your result is within ε of $\pi = 3.1415926 \dots$

(c) Make a plot of ε versus the number N of required terms in your algorithm on a log-log scale. What is the slope of the line that the resulting plot approaches as N increases? Plot the actual error as a function of N on the same axes, you may treat the stored value of π as “exact.” How does the actual error compare to the estimated threshold?

Solution. (a) Our algorithm is given in the file “problem1.m” but also below as it is simple:

```
function approximatePi = computePiPlotError(error)
    p = 0;
    sum = 0;
    index = 0;
    pApproximated = [];
    indexedError = [];
    while (abs(pi - p) > error)
        sum = sum + (-1)^index * 1/(2*index + 1);
        p = 4*sum;
        index = index + 1;
        pApproximated(index) = p;
        err = abs(pi - p);
        indexedError(index) = err;
```

```

end
approximatePi = p;
loglog(1:index,indexedError,'--k')
xlabel('Iteration Index n');
ylabel('Error');
title('Error in Pi Approximation');
end

format long
approx_pi = computePiPlotError(5e-3)
pi - approx_pi

```

Note, an upper bound on ε should be $|pi - 4| \approx 0.85840$.

(b) When we run this function we get the following output from matlab:

$$p = 3.136592684838816$$

$$\varepsilon = 0.004999968750977$$

(c) When we plot ε versus the number of iterations of our algorithm to try to find π we get: I'm not exactly sure what we're going for here with the "actual error". The only thing

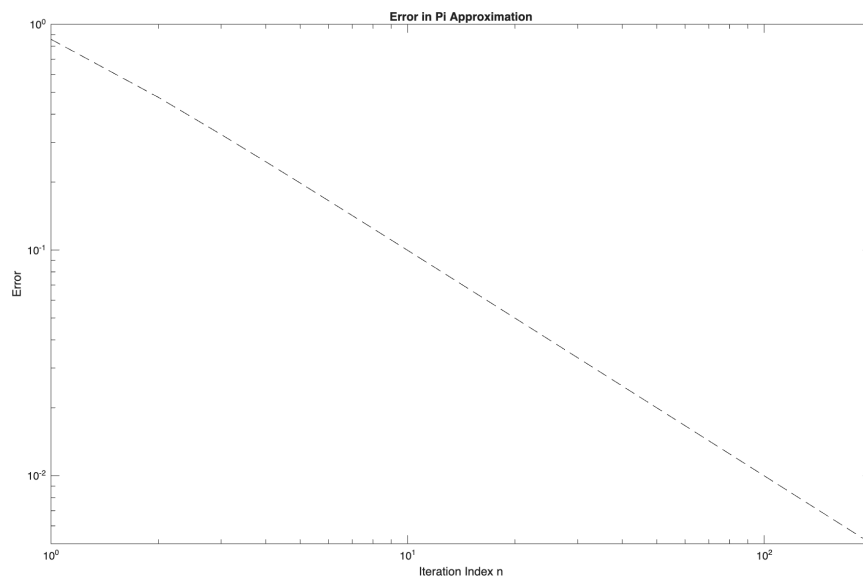


FIGURE 1. Log-Log plot ε versus N

that I can come up with is considering the remainder of the tail of the summation after the

n^{th} iteration.

$$\text{err}_n^{\text{actual}} = 4 \sum_{k=n+1}^{\infty} \frac{(-1)^k}{2k+1}$$

I suppose we could pick some huge k , say 10^{12} and iterate from our current n to there to get the actual error. But to do so I would end up using the symbolic computation capabilities in matlab. I've added a "problem1b.m" in the directory but the computational capacity on the "symsum" function is insufficient for the problem.

□

2. Numerically determine which of the following limits approaches 1 faster. Then confirm the numerical evidence by determining the rate of convergences of each sequence

$$\lim_{n \rightarrow 0} \frac{\sin(x^2)}{x^2} \quad \text{versus} \quad \lim_{n \rightarrow 0} \frac{(\sin x)^2}{x^2}.$$

Solution. We'll just run a table here for $\{10^0, 10^{-1}, 10^{-2}, \dots, 10^{-9}\}$ to look at convergence of the functions listed above. The code for this problem is minimal command line work using vectors with the scalar versions of the sin and arithmetic functions.

n	$\sin(x^2)/x^2$	$(\sin x)^2/x^2$
10^0	-0.005063656411098	0.002959589690933
10^{-1}	0.841470984807897	0.708073418273571
10^{-2}	0.999983333416666	0.996671107937918
10^{-3}	0.999999998333333	0.999966667111108
10^{-4}	0.99999999999833	0.999999666666711
10^{-5}	1.000000000000000	0.999999996666667
10^{-6}	1.000000000000000	0.999999999966667
10^{-7}	1.000000000000000	0.999999999996667
10^{-8}	1.000000000000000	0.99999999999997
10^{-9}	1.000000000000000	1.000000000000000

Temp

□

3. (a) Consider the function $f_c(x) = x^2 - 12x + c$. What are the zeros of f_c with $c = 35$? What are the zeros of f_c with the slight constant shift $c = 35 - 10^{-2}$? Relative to the size of change in the constant term c , how big is the change in the zeros of the function?

(b) consider the function $g_c(x) = x^2 - 10x + c$. What are the zeros of g_c with $c = 25$? What are the zeros of g_c with the slightest constant shift $c = 25 - 10^{-2}$? Relative to the size of change in the constant term, how big is the change in the zeros of the function?

(c) Compare the nature of the roots of f and g . How might this contribute to the relative changes in the zeros for each function?

Solution. Problem 3

□

4. Identify the potential roundoff error problems in the following algorithm for calculating the roots of the quadratic equation $ax^2 + bx + c = 0$.

GIVEN: real coefficients a, b, c

STEP 1: calculate $disc = \sqrt{b^2 - 4ac}$

STEP 2: calculate $root1 = -2c/(b + disc)$

STEP 3: calculate $root2 = (-b/a) - root1$

OUTPUT: $root1$ and $root2$

Note that this algorithm uses the fact that the sum of the roots of $ax^2 + bx + c = 0$ is equal to $-b/a$.

Solution. Problem 4 □

5. (a) Plot the function $f(x) = \ln(1+x) - \cos(x) - x + 1$ over the interval $-5 \times 10^{-6} \leq x \leq 5 \times 10^6$ using 1001 uniformly spaced points in x .

(b) Reformulate f to avoid the cancellation error, and then repeat part (a).

(c) Describe the issues with the original formulation in part (a) and how they were resolved with the reformulation in part (b).

Solution. Problem 5 □

6. (a) Determine the third-degree Taylor polynomial associated remainder term for the function $f(x) = \sqrt{1+x}$. Use $x_0 = 0$.

(b) Using the result of part (a), approximate $\sqrt{1.5}$, and compute the theoretical error bound associated with this approximation. Compare the theoretical error bound with the actual error.

(c) Compute the following limit and determine the corresponding rate of convergence:

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1 - \frac{1}{2}x}{x^2}$$

Solution. Problem 6 □

7. Consider the linear system $Ax = b$ with $A = \begin{bmatrix} 3.4 & 2.8 \\ 8.0 & 6.6 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, and $b = \begin{bmatrix} 3.4 \\ 8.0 \end{bmatrix}$.

(a) Determine the analytic solution by hand. (How do we know that it is unique, i.e. that the problem is well-posed?)

(b) Now solve the system with $b = \begin{bmatrix} 3.41 \\ 8.0 \end{bmatrix}$. You may use your computer.

(c) Given the difference in solutions for parts (a) and (b), comment on the conditioning of the problem. What is $\det[A]$ and for what values with the problem be ill-posed?

Solution. Problem 7 □